

Scientific terms

1- Magnetic field:

It is the region in which the magnetic force appears through it.

2- The magnetic flux (Φ):

It is the total number of the magnetic field lines that pass perpendicular to surface area.

3- Magnetic flux density at a point (B):

*It is the magnetic flux that passes through normal a unit area around this point

*It is the magnetic force that acts on a straight wire of length 1m , carrying electric current 1A placed normal to a magnetic field.

4- Tesla:

*A magnetic force of 1N acts on a wire carrying current 1A , placed normal to a magnetic field

* the magnetic flux lines that passes normal to a unit area = 1 wb

5- The magnetic field:

It is the region that appears the magnetic force though it

6- Magnetic permeability μ :

It is the ability of the medium to allow passing the magnetic field through it.

7- Neutral point:

*It is the point that the magnetic flux density vanishes through it ($B_t = \text{zero}$).

8. Electromagnetic induction:

It is a phenomenon of producing induced electromotive force and induced electric current in a conductor due to the variation of magnetic flux lines that is linked by a conductor

9- Lenz's rule:

The induced current that passes in a conductor opposes the variation that producing it

10- Faraday's law:

The induced electromotive force in a conductor is directly proportional with the rate of variation of magnetic flux lines linked by a conductor ($\Delta\Phi/\Delta t$) and is directly proportional with the no.of turns of the conductor that cut the magnetic flux (N)

11- The mutual induction between two coils:

It is the electromagnetic induction that is produced in 2nd coil due to the variation of electric current in 1st coil. the induced current opposes the variation of the electric in 1st coil.

12- The coefficient of mutual induction between coil (M):

It is the induced electromotive force in a coil due to the variation of electric current in other coil by 1A in 1sec.

13- The self-induction:

It's electromagnetic induction that produced in a coil due to the variation of electric current in same coil to oppose this variation

14- The coefficient of self induction in a coil (L):

It's induced EMF in a coil due to the variation of electric current in the same coil by 1A in 1sec

15- Henry:

It's is the mutual induction between 2 coil to produce EMF = 1V in a coil when , the electric current changes in other coil by rate 1 A/s.

16- Eddy currents:

They are induced currents that are generated in a metallic block as a closed path due to the variation of a magnetic flux and generate heat.

17- AC generator:

It is a device that is used to convert the kinetic energy into electric energy

18- Alternating current (AC):

*It's electric current changes its direction and its intensity periodically

19- The transformer:

It is a device that is used to step up or step down in the voltage

20- The efficiency of electric transformer:

*It's the ratio between the electric power in secondary coil (OUT PUT) to the electric power in primary coil (IN PUT)

21- DC motor:

It is a device that is used to convert the electric energy into kinetic energy(torque)

22- Black body:

it is the body that absorbs all radiation falling on it (perfect absorber) and reemits again (perfect emitter)

23- Plank's curve:

it is a graphical relation between the intensity of continuous radiations that emits from any hot bodies and the wavelengths of these radiations

24- Plank's distribution:

It is the ditribution between The intensity of radiations that emits from hot bodies and the wavelength.

25- Wien's law:

the wavelength (λ_m) accompanying the maximum intensity of radiation is inversely proportional with the kelvin temperature of hot bodies.

26- Night vision system:

imaging the moving objects in the dark due to the infrared radiations that emit from it.

27- Remote sensing:

it is a technology that is used to indicate the nature resources at the surface of earth by analyzing the infrared radiations of the earth.

28- Surface potential barrier:

it is the attractive force between the positive ions in the metallic surface and the free electrons

29- Thermal emission (thermionic effect):

it is the emission (liberation – escaping) of free electron from the metallic surface by heat.

30- Photoelectric effect:

it is the emission of free electrons from the metallic surface by light falling with frequency is greater than the critical frequency.

31- Work function:

it is the minimum energy required to liberate free electrons from the metallic surface without gaining energy.

32- Critical frequency:

it is the minimum frequency of the incident light on a metallic surface to liberate electrons from this surface.

33- Photoelectric current:

it is the emitted electrons from the metallic surface due to falling light of frequency greater than the critical frequency.

34- Compton's effect:

when a photon of x – rays or γ rays collides with an electron at rest, its frequency decreases and its direction changes and the velocity of electron increases and its direction changes.

35- Photon:

it is a quantum of energy ($h\nu$) has no charge and has particle nature.

36- electron:

it is a particle with –ve charge and has wave nature.